

ECON100B - SECTION 22

Government & PSET 4

Felix Zhang

ANNOUNCEMENTS/AGENDA

Announcements

- PSET 4 released, due Friday 5/1

Agenda

- Government
- PSET 4

Question of the Day

Problem Set 4, Q1.d

- (d) [(3 points) Suppose inflation were to rise unexpectedly. In credit markets — think of banks and their customers — whom would this hurt, if anyone? Whom would this help, if anyone? Explain.

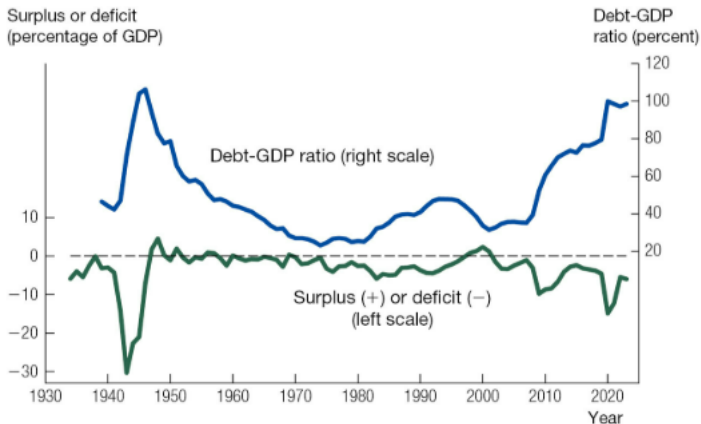
Government



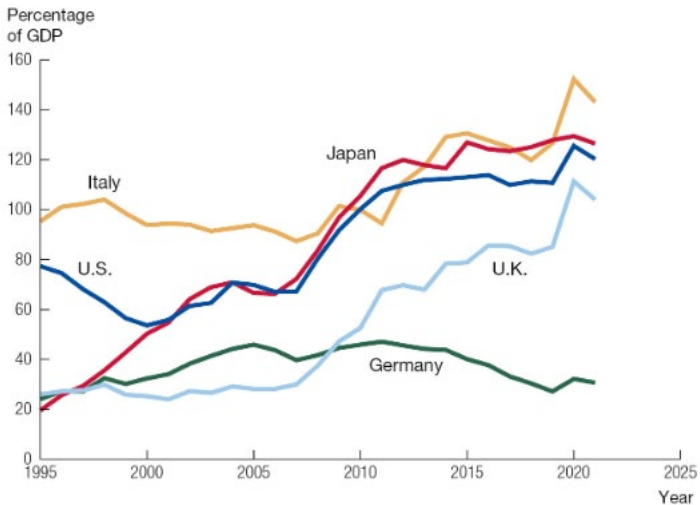
The government in the macroeconomy

- Federal spending typically hovers between 20-25% of the GDP so a huge component of the total macroeconomy
- Biggest categories of spending include healthcare, national defense, social security, and income security, and interest payments on debt (!)
- Because government spending in recent history has almost always been higher than tax revenues, the government uses debt financing to make up for the gap
- We can use the debt-GDP ratio to calculate the amount of debt owed vs the GDP of a country
- In the U.S., this has risen from 35% to 100% in the past 20 years
- (Almost) anyone can buy U.S. debt. Individual investors, institutional investors, foreigners, etc.

Debt GDP ratio



Debt-GDP Ratios around the World



The Government Budget Constraint

- The government must have a “balanced” budget, leading us to the following budget constraint:

$$\underbrace{G_t + Tr_t + iB_t}_{\text{uses}} = \underbrace{T_t + \Delta B_{t+1} + \Delta M_{t+1}}_{\text{sources}}.$$

- Uses: Government spending, transfer payments, and interest payments
- Sources: Tax revenue, new borrowing bonds, and new money changing the money supply (assume 0 given fiscal-monetary independence)

Simplifying the Budget Constraint

$$B_{t+1} = (1 + i)B_t + \underbrace{G_t - T_t}_{\text{deficit}}$$

- The stock of debt next year equals:
- The stock of debt this year, including interest payments
- The change in the government deficit based on spending minus transfer payments

Government Practice

1. What is the debt to GDP ratio in the United States? Has it been growing over time and how does it compare to other parts of the world?
2. Suppose I have \$5T in government spending and \$4.5T in tax revenue in a given year. Also suppose that I have a current debt of \$4T with a 5% interest rate. What will my debt be next year?

Government Practice Solutions

1. What is the debt to GDP ratio in the United States? Has it been growing over time and how does it compare to other parts of the world?
2. \$4.7T of debt. Solve via $\$4T * (1.05) + \$0.5T$

Why Should the Government Spend?

Three Big Reasons

1. Wealth Redistribution
 2. Provide Non-market Goods + Services
 3. Step in to help the economy when the Central Bank/Monetary Policy becomes ineffective
- Government spending also directly positively impacts the GDP!
 - We're less certain about the positive economic impacts of transfer payments (since we're just moving money around), but there are a lot of equity considerations around transfer payments

What are Concerns about Government Spending?

- Eventually, the government must pay back all its debts.
 - The good news: the government “doesn’t die” so it can keep taking out more and more debt
 - The bad news: there probably is a limit to how much debt we can/should take out, before we see too many negative impacts
- Government spending crowds out investment spending
 - When we put our money into government bonds to pay off debt, it means that we aren’t putting that money into other forms of investment

Government Spending and Investment

How do we see this? Start with income accounting:

$$Y = C + I + G + EX - IM$$

Now rearrange, putting investment, I , on one side:

$$I = Y - C - G - EX + IM$$

Taxes, T , are subtractions from your income. So let's add and subtract T from the right hand side, and flip EX and IM:

$$I = \underbrace{Y - C - T}_{\text{Income minus consumption minus taxes is what you have left for private saving}} + \underbrace{T - G}_{\text{Taxes minus spending equals public saving}} + \underbrace{IM - EX}_{\text{Imports minus exports is foreign saving (think: imported investment goods)}}$$

Income minus consumption
minus taxes is what you
have left for **private
saving**

Taxes minus
spending equals
public saving

Imports minus exports
is **foreign saving**
(think: imported
investment goods)

PSET 4, Q2

2. [20 points total] Saved by how the Government always asks about Investment

Recall that national income accounting requires that

$$Y = C + I + G + EX - IM \quad (2.1)$$

Total spending — consumption, investment, government, and net exports — sums to GDP, or national production, which is also equal to national income. Throughout this problem, it will help you to think of Y as income and fixed in the short run, even though in reality it is not.

- (a) [3 points] Leave $EX - IM$ on the right-hand side and subtract everything else except Y from both sides.

PSET 4, Q2

- (b) [3 points] On the left-hand side, add and subtract taxes, T . By bracketing with parentheses and labeling, show: (i) private saving, (ii) public or government saving, and (iii) investment.

PSET 4, Q2

Recall that in ECON 100B, we typically assume that total taxes T and income tax rates do not change working or consuming behavior. In other words, increases in taxes T directly reduces after-tax income, but we assume that higher tax rates do not change labor supply or income, and that they also do not change consumption. (For small changes, this is probably realistic.)

- (c) [3 points] Under these assumptions, what are the effects, if any, of an increase in taxes, T , according to your answer to part (b)? What about a reduction in taxes? Explain.

PSET 4, Q2

- (d) [3 points] Make an argument for why the government should run a budget surplus by reducing G below T . Describe how that might increase a component of GDP, using equation (2.1) and your answer to part (b). (I can think of two; choose one.)
- (e) [3 points] Now make a macroeconomic argument for why the government should run a budget deficit, by raising G above T . Describe how that could be good for GDP, using equation (2.1) and your answer to part (b) as need be.

PSET 4, Q2

Suppose I told you that research in macroeconomics¹ suggests that when the government spends an additional $\Delta G = \$1.00$ on national defense, it appears that GDP actually rises on net by about $\Delta Y = \$0.80$ as a result. Why? Government spending partially *crowds out* other components of GDP, because it requires either higher taxes or higher borrowing or both.

- (f) [5 points] Discuss this in light of your previous answers. Does this perspective suggest that one of the two arguments in parts (d) and (e) might be more wise? Explain fully.